

Table 5.3: Functional Requirements

ID	Requirement	Description	Priority
FR-01	User Registration	System shall allow new users to register with email and password, selecting their role (Farmer, Agent, Admin)	High
FR-02	User Authentication	System shall authenticate users via Firebase Authentication with email/password credentials	High
FR-03	Password Reset	System shall allow users to reset their password via email verification link	Medium
FR-04	Waste Listing Creation	Farmers shall be able to create waste listings with title, category, quantity, price, location, photos, and description	High
FR-05	Listing Discovery	Agents shall be able to browse all open waste listings sorted by creation date	High
FR-06	Listing Claiming	Agents shall be able to claim open listings, changing status to ASSIGNED	High
FR-07	Pickup Workflow	Agents shall transition listings through ASSIGNED → IN_TRANSIT → DELIVERED states	High
FR-08	Listing Cancellation	Farmers shall be able to cancel their own OPEN listings	Medium
FR-09	Payment Processing	System shall integrate Razorpay for processing payments between buyers and sellers	High
FR-10	Carbon Credit Calculation	System shall calculate carbon offsets based on waste type and quantity using established emission factors	Medium
FR-11	Rewards System	System shall track user points, streaks, and milestones based on platform activity	Low
FR-12	Photo Upload	Users shall be able to upload waste photos via Cloudinary integration during listing creation	Medium